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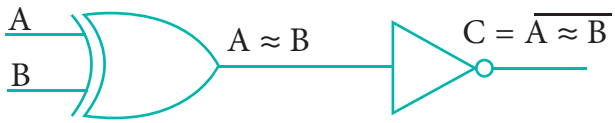

Fig. 2.17 Logic Circuit of XNOR Gate

The output of the XNOR is NOT of XOR

$$C = \overline{A \oplus B}$$

$$= \overline{A} \cdot B + A \cdot \overline{B}$$

$$= AB + \overline{A}\ \overline{B}$$

(Using De Morgan's Theorem)

In boolean algebra, $\odot$ or "included dot" stands for the XNOR.

Therefore, $C = A \odot B$

The logical symbol is

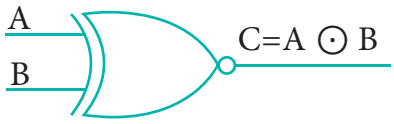

Fig. 2.18 Logic Symbol of XNOR Gate

The truth table for XNOR Gate is

Input		Output
A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

Table 2.14 Truth Table for XNOR Gate

Using combination of logic gates, complex operations can be performed. In theory, there is no limit to the number of gates that can be arranged together in a single device. But in practice, there is a limit to the number of gates that can be packed into a given physical space. Arrays of logic gates are found in digital integrated circuits.